Yasitha Warahena Liyanage — Curriculum Vitae

Contact Information

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Main Interests

Statistical Signal Processing, Machine Learning, Cloud Computing, Optimization Theory.

Education

University at Albany, **State University of New York (SUNY)**, Albany, NY
PhD, Electrical and Computer Engineering — **January 2022**
G.P.A.: 3.98 out of 4.
Dissertation Title: Dynamic Instance–Wise Decision–Making for Machine Learning.
Advisor: Dr. Daphney-Stavroula Zois.

University of Peradeniya, Sri Lanka
BSc, Electrical and Electronic Engineering — **October 2016**
G.P.A.: 3.95 out of 4. Standing: 3rd out of 110.

Experience

Senior Data and Applied Scientist — **March 2024–present**
Microsoft Corporation, Redmond, WA
- Lead the architecture and deployment of large–scale, real–time anomaly detection systems for cloud infrastructure, reducing time–to–detection of incidents and driving high availability.
- Enable accelerated root–cause attribution and early mitigation via the integration of causal inference methodologies across large–scale telemetry datasets.

Data and Applied Scientist–II — **January 2022–March 2024**
Microsoft Corporation, Redmond, WA
- Engineered high–precision machine learning models for Azure cloud service performance anomaly detection, improving signal quality and eliminating false alerts to reduce operational fatigue.
- Expanded distributed system observability across global cloud layers by translating complex telemetry insights into highly scalable, production–grade monitoring frameworks.

Applied Scientist Intern — **Summer 2020, Summer 2021**
Amazon Capital Services, Inc, Seattle, WA
- Secured and scaled the automated underwriting of new seller loan portfolios by designing and deploying production–level credit risk models for the Amazon Seller lending platform.

Graduate Research Assistant — **August 2017–December 2021**
Department of Electrical & Computer Engineering, **University at Albany**, **State University of New York (SUNY)**, Albany, NY
- Designed algorithms to perform dynamic instance–wise decision–making.
- Designed algorithms to optimally detect the time and geographic location of accidents in near–real–time in a road segment equipped with spatially distributed speed sensors.

Advisors: Dr. Daphney–Stavroula Zois and Dr. Charalampos Chelmis.

Graduate Research Assistant — **November 2016–July 2017**
Department of Electrical and Electronic Engineering, **University of Peradeniya**, Sri Lanka
- Developed a real–time Non–Intrusive Load Monitoring (NILM) system using uncorrelated spectral components of the active power consumption signal based on Karhunen–Loéve expansion.
- Implemented a nonlinear controller for a standard twin rotor multi–input multi–output system using state feedback linearizing techniques.

Advisors: Dr. Janaka Ekanayake, Dr. Roshan Godaliyadda, Dr. Parakrama Ekanayake and Dr. Lilantha Samaranayake.

Teaching Experience

Teaching Assistant — **November 2016–July 2017**
Department of Electrical and Electronic Engineering, **University of Peradeniya**, Sri Lanka

- EE 501: Advanced Control Systems
- EE 539: Nonlinear and Multi-variable Systems

Awards

Microsoft Patent Award, Microsoft Corporation	**2025**
Azure Core Champ Award, Microsoft Corporation	**2023**
Distinguished Doctoral Dissertation Award, University at Albany	**2022**
Graduate Student Association Grant Award, University at Albany	**2020**
President's Award for Scientific Research, Sri Lanka	**2019**
Full Colors, Colors Awarding Ceremony, University of Peradeniya, Sri Lanka	**2016**

Patents & Publications

Patents

1. **Y. Liyanage**, K. Hsieh, M. Chintalapati, V. Ramesh, S. Zuehlsdorff, "*Real–Time Detection of Latency Anomalies Affecting Cloud Production Workloads*," Reference no. 503487–US01 (Filled to USPTO) Jan 2025.

Selected Articles & Journals

4. **Y. Liyanage**, D.-S. Zois, C. Chelmis, "*Dynamic Instance–wise Classification in Correlated Feature Spaces*," IEEE Transactions on Artificial Intelligence, vol. 2, no. 6, pp. 537–548, December 2021. [pdf]

3. **Y. Liyanage**, D.-S. Zois, C. Chelmis, "*Dynamic Instance–wise Joint Feature Selection and Classification*," IEEE Transactions on Artificial Intelligence, vol. 2, no. 2, pp. 169–184, April 2021. [pdf]

2. **Y. Liyanage**, D.-S. Zois, C. Chelmis, "*Near Real–Time Freeway Accident Detection*," IEEE Transactions on Intelligent Transportation Systems, October 2020. [pdf]

1. C. Dinesh, S. Welikala, **Y. Liyanage**, M.P.B. Ekanayake, R.I. Godaliyadda, J. Ekanayake, "*Non–Intrusive Load Monitoring Under Residential Solar Power Influx*," Elsevier Journal of Applied Energy, vol. 205, pp. 1068–1080, November 2017. [pdf]

Selected Conferences & Workshops

16. **Y. Liyanage**, X. Liu, K. Hsieh, M. Chintalapati, "*Causal Attribution of Azure Control-Plane Anomalies*". 2026 Machine Learning, AI & Data Science Conference (MLADS 2026), Redmond, WA, June 08–11, 2026.

15. **Y. Liyanage**, D.-S. Zois, "*Interpretability in the Context of Sequential Cost–Sensitive Feature Acquisition*". 48th IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP 2023), Rhode Island, Greece, June 04–10, 2023. [pdf]

14. X. Yan, K. Hsieh, **Y. Liyanage**, M. Ma, M. Chintalapati, Q. Lin, Y. Dang, D. Zhang, "*Aegis: Attribution of Control Plane Change Impact across Layers and Components for Cloud Systems*". 45th International Conference on Software Engineering: Software Engineering in Practice (ICSE-SEIP 2023), Melbourne, Australia, May 14–20, 2023. [pdf]

13. **Y. Liyanage**, D.-S. Zois, "*Optimum Feature Ordering For Dynamic Instance–wise Joint Feature Selection and Classification*". 46th IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP 2021), Toronto, Ontario, Canada, June 06–11, 2021. [pdf]

12. S. Ekanayake, **Y. Liyanage**, D.-S. Zois, "*Dynamic Feature Selection for Classification in Structured Environments*". 55th Asilomar Conference on Signals, Systems, and Computers (ACSSC 2021), Pacific Grove, CA, October 31–November 03, 2021. [pdf]

11. I. Nazar,**Y. Liyanage**, D.-S. Zois, C. Chelmis, "*Sequential Heterogeneous Feature Selection for Multi-class Classification: Application in Government 2.0*", IEEE International Workshop on Machine Learning for Signal Processing (MLSP), Aalto University, Espoo, Finland, September 21–24, 2020. [pdf]

10. **Y. Liyanage**, D.-S. Zois, C. Chelmis, "*On–the–fly Feature Selection and Classification with Application to Civic Engagement Platforms*". 45th IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP 2020), Barcelona, Spain, May 04–08, 2020. [pdf]
9. I. Nazar, **Y. Liyanage**, D.-S. Zois, C. Chelmis, "*Automated Optimal Online Civil Issue Classification using Multiple Feature Sets*," Asilomar Conference on Signals, Systems, and Computers (ACSSC 2019), Pacific Grove, CA, November 03–09, 2019. [pdf]
8. **Y. Liyanage**, D.-S. Zois, C. Chelmis, M. Yao,"*Robust Freeway Accident Detection: A Two-Stage Approach*," 44th IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP 2019), Brighton, UK, May 12–17, 2019. [pdf]
7. **Y. Liyanage**, D.-S. Zois, C. Chelmis, "*Automating The Classification Of Urban Issue Reports: An Optimal Stopping Approach*," 44th IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP 2019), Brighton, UK, May 12–17, 2019. [pdf]
6. **Y. Liyanage**, C. Chelmis, D.-S. Zois, "*A Hierarchical Framework for Timely Freeway Accident Detection and Localization*," IEEE International Conference on Big Data (BigData 2018), Seattle, WA, December 10–13, 2018. [pdf]
5. **Y. Liyanage**, D.-S. Zois, C. Chelmis, "*Quickest Freeway Accident Detection Under Unknown Post–Accident Conditions*," 6th IEEE Global Conference on Signal and Information Processing (GlobalSIP 2018), Anaheim, CA, November 26–29, 2018. [pdf]
4. **Y. Liyanage**, M. Yao, C. Yong, D.-S. Zois, C. Chelmis, "*What matters the most? Optimal Quick Classification of Urban Issue Reports by Importance*," 6th IEEE Global Conference on Signal and Information Processing (GlobalSIP 2018), Anaheim, CA, November 26–29, 2018. [pdf]
3. **Y. Liyanage**, D.-S. Zois, C. Chelmis, "*Optimal Sequential Detection of Freeway Accidents*," Asilomar Conference on Signals, Systems, and Computers (ACSSC 2018), Pacific Grove, CA, October 28–31, 2018. [pdf]
2. J. Wijekoon, **Y. Liyanage**, S. Welikala, L. Samaranayake, "*Yaw and Pitch Control of a Twin Rotor MIMO System Using a Nonlinear based Controller*," IEEE International Conference on Industrial and Information Systems (ICIIS 2017), Peradeniya, Sri Lanka, December, 2017. [pdf]
1. **Y. Liyanage**, S. Welikala, C. Dinesh, M. P. B. Ekanayake, R. I. Godaliyadda, J. Ekanayake, "*Real-time non-intrusive appliance load monitoring under supply voltage fluctuations*," IEEE International Conference on Advances in ICT for Emerging Regions (ICTer 2017), Colombo, Sri Lanka, September, 2017. [pdf]

PhD Thesis

1. **Y. Liyanage**, "*Dynamic Instance–Wise Decision–Making for Machine Learning*," Albany, NY, January, 2022. [pdf]

Technical Skills

- Python, KQL, SQL, Matlab, AWS, Azure
- Linux, macOS, Windows

Professional Activities / Memberships

- **TCP–Chair:** International Conference On Industrial And Information Systems (ICIIS–2025)
- **Reviewer:** Springer Nature, Machine Learning
- **Reviewer:** IEEE Transactions on Industrial Informatics
- **Reviewer:** Machine Learning, AI & Data Science Conference (MLADS–2023)
- **Program Committee:** AAAI Conference on Artificial Intelligence 2021 (AAAI–21)
- **Member:** IEEE, IEEE Signal Processing Society

Languages

- **English**
- **Sinhala**